Capstone Project: Student Guide to Research Topics, Data Combinations, and GitHub Collaboration

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Purpose

This guide helps you:

1. Choose a compelling research topic (or refine the default REIT question)
2. Identify effective data combinations for your analysis
3. Set up and use a shared GitHub repository as your team's source of truth

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Part 1: Choosing Your Research Topic

The Core Framework: Outcome Sensitivity Analysis

Every capstone project follows this template:

> *"[Decision-Maker] needs analysis of [Outcome Variable] sensitivity to [Driver Variable], with focus on [Heterogeneity]."*

Key Components:

1. **Decision-Maker:** Who cares about this? (Investment Committee, Policy Analysts, Risk Committee, etc.)
2. **Outcome Variable:** What are we trying to understand? (returns, growth, volatility, prices)
3. **Driver Variable:** What influences it? (interest rates, policy shocks, economic indicators)
4. **Heterogeneity:** Does the effect vary? (across sectors, regions, time periods, firm types)

Option 1: Default REIT Track (Recommended)

Research Prompt:

> "You are a Junior Quantitative Analyst at a real estate investment firm. The Investment Committee is concerned about REIT return sensitivity across sectors following the 2022-2023 Federal Reserve interest rate hikes. Build a data pipeline to analyze historical sensitivity of REIT returns to interest rates and economic factors, and estimate the causal impact of market shocks on different REIT sectors."

Dataset Structure:

- **Primary:** REIT Master dataset (500+ REITs, monthly, 2004-2024)
  - **Outcome variables:** `ret` (monthly return), `ret_excess` (excess return over risk-free rate), `volatility` (realized volatility), `sharpe` (risk-adjusted return)
  - **Entity characteristics:** `permno` (REIT ID), `sector` (property type), `mcap` (market cap), `price`, `volume`, `leverage` (debt ratio), `ffo` (funds from operations), `dividend_yield`

- **Factor exposures:** `beta_market`, `beta_size`, `beta_value` (from REIT factor model)

- **Supplementary data (ADD MORE—richer data = stronger analysis):**
  - **FRED Macroeconomic (10-15 series recommended):**
    - Interest rates: `FEDFUNDS`, `MORTGAGE30US`, `DGS10` (10-year Treasury), `TERMSPREAD` (yield curve slope)
    - Inflation: `CPIAUCSL`, `PCEPI`, `CPILFESL` (core inflation)
    - Labor: `UNRATE`, `PAYEMS` (nonfarm payroll), `AWHMAN` (average hours worked)
    - Output: `INDPRO` (industrial production), `GDP`, `RSXFS` (retail sales)
    - Credit: `MORTGAGE30US`, `DEXCHUS` (USD/CNY), `VIXCLS` (market volatility)
  - **Real estate specific (Open Dataset Catalog or FRED):**
    - Housing price indices (national, regional): `CSUSHPISA` (Case-Shiller)
    - Construction spending: `TLRESCONS`
    - Mortgage delinquency rates
    - Commercial vacancy rates by metro
  - **Policy events (hand-coded or scraped):**
    - Fed rate hike announcements (date + magnitude)
    - COVID-19 lockdown dates (state-level)
    - Major fiscal stimulus events

Why choose this?

- Aligned with weekly assignments (no translation needed)
- Rich dataset with 20+ variables available
- Peer support and shared debugging
- Well-documented with examples
- Instructor is expert in this domain

Data Richness Principle: More variables give you robustness checks, alternative specifications, and richer stories. Don't limit yourself to 3-4 variables—aim for 10-15 supplementary series.

Research Question Examples (REIT Track):

1. **Interest Rate Sensitivity by Sector (Multi-Outcome)**
   - *Primary question:* "Do Retail REITs respond differently to interest rate changes than Industrial REITs?"
   - *Outcomes to test:* Returns, volatility, trading volume, Sharpe ratios
   - *Heterogeneity:* Sector (Retail vs. Office vs. Industrial vs. Residential vs. Healthcare)
   - *Supplementary data:* FEDFUNDS, MORTGAGE30US, DGS10, TERMSPREAD, VIXCLS, GDP, UNRATE (8+ series)
   - *Why multiple outcomes?* Returns tell you price impact; volatility tells you risk response; volume tells you liquidity effects

2. **Leverage Channel Analysis (Mechanism Testing)**
   - *Question:* "Is interest rate sensitivity amplified for highly leveraged REITs?"
   - *Outcomes:* Returns, excess returns, default risk proxies
   - *Heterogeneity:* Leverage quartiles + size quartiles (interaction)

- *Supplementary data:* FEDFUNDS, MORTGAGE30US, credit spreads (BAA-AAA), VIXCLS, RSXFS (consumer spending)
- *Mechanism:* High leverage → higher interest expense → lower cash flows → bigger return drop

3. **COVID-19 Shock Analysis (Event Study)**

   - *Question:* "Which REIT sectors experienced the largest shocks during COVID-19, and did they recover differently?"
   - *Outcomes:* Cumulative returns, volatility, dividend cuts
   - *Heterogeneity:* Sector + Pre/During/Post COVID periods
   - *Supplementary data:* UNRATE, retail sales, mobility indices (Google Mobility), state lockdown dates, fiscal stimulus amounts
   - *Policy implication:* Did government support cushion certain sectors?

4. **Geographic Exposure and Housing Markets**

   - *Question:* "Do REITs with coastal exposure outperform inland REITs during housing booms?"
   - *Outcomes:* Returns, FFO growth, occupancy rates
   - *Heterogeneity:* Coastal vs. Inland + Urban vs. Suburban
   - *Supplementary data:* Case-Shiller indices (20 cities), median household income by metro, construction spending, mortgage originations
   - *Why geography?* Housing markets are regional—national averages hide heterogeneity

5. **Multiple Outcome Robustness Example**

   - *Test same hypothesis across 3 outcomes:*
     - Returns → price effect
     - Volatility → risk effect
     - Dividend yield → cash flow effect
   - *Why?* If all three respond similarly, your finding is robust. If only returns respond, the effect might be speculative (not fundamental).

Option 2: Alternative Dataset from Open Dataset Catalog

Requires: Formal proposal + instructor approval by **Friday, Week 4 (Feb 13)**

Proposal Checklist:

- ☐ Dataset meets minimums (50+ entities OR 5+ years time series)
- ☐ Clear research question using the framework above
- ☐ Identified outcome + driver + heterogeneity dimension
- ☐ Access method documented (API, download link, etc.)
- ☐ Reviewed `Dataset-Translation-Examples.md` to understand adaptation needs

Example Alternative Research Questions:

Cryptocurrency Track (Rich Data Version)

Research Prompt:

> "Regulatory Committee needs analysis of cryptocurrency return sensitivity to SEC enforcement actions, with focus on DeFi vs. CEX token differences."

- **Outcomes (test multiple):**
  - Daily returns (`return_pct`)
  - Realized volatility (30-day rolling)
  - Trading volume (liquidity proxy)
  - Bid-ask spreads (market quality)
- **Primary driver:** Regulatory severity index (1-10 scale from SEC announcements)
- **Entity characteristics:**
  - Token type (DeFi/CEX/NFT/Layer-1/Layer-2)
  - Market cap category
  - Age (days since launch)
  - Exchange listings (centralized vs. DEX only)
  - Total value locked (TVL) for DeFi protocols
- **Supplementary data (aim for 10-15 series):**
  - **Crypto-specific:** Bitcoin return (market benchmark), Ethereum gas prices, stablecoin dominance, DeFi TVL aggregate
  - **Macro:** VIX (risk aversion), FEDFUNDS, DXY (dollar index), gold prices
  - **Policy:** SEC enforcement action dates, Congressional hearing dates, country-level regulatory changes

Why this works: Policy shock identification, multiple outcomes show robustness, rich cross-sectional variation

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Housing Market Track (Comprehensive)

Research Prompt:

> "Federal Reserve economists need analysis of regional home price sensitivity to mortgage rate changes, with focus on urban vs. suburban dynamics and income effects."

- **Outcomes (test multiple):**
  - Home price growth (year-over-year)
  - Rent growth
  - Sales volume (transaction activity)
  - Price-to-rent ratio (valuation)
- **Primary driver:** Mortgage rates (MORTGAGE30US)
- **Entity characteristics:**
  - MSA/county identifier
  - Urban/suburban classification
  - Coastal vs. inland

- Median household income quartile
- Housing supply elasticity (Saiz index)
- Population growth rate

- **Supplementary data (10-20 series recommended):**
  - **Housing-specific:** Case-Shiller indices (20 cities), construction permits, housing starts, inventory (months of supply), foreclosure rates
  - **Economic:** UNRATE (national + regional), median income, GDPC1, retail sales
  - **Financial:** FEDFUNDS, 10-year Treasury, mortgage spread (MORTGAGE30US - DGS10), credit availability index
  - **Demographics:** Migration flows, millennial homeownership rates, investor share of purchases

Why this works: Regional heterogeneity, multiple transmission channels (affordability, supply, demand), established empirical literature

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Macroeconomic Track (Multi-Country Panel)

Research Prompt:

> "IMF policy analysts need analysis of employment growth sensitivity to monetary policy tightening, with focus on institutional quality and labor market flexibility."

- **Outcomes (test multiple):**
  - Employment growth (total, by sector)
  - Unemployment rate changes
  - Labor force participation
  - Real wage growth

- **Primary driver:** Central bank policy rate changes (Fed, ECB, BoE, BoJ)

- **Entity characteristics:**
  - Country identifier
  - OECD vs. Emerging market
  - Labor market flexibility index (OECD EPL)
  - Unionization rate
  - GDP per capita

- **Supplementary data (15-25 series for robustness):**
  - **Monetary:** Policy rates (multiple countries), inflation, exchange rates, yield curves
  - **Fiscal:** Government spending (% GDP), debt-to-GDP, fiscal stimulus episodes
  - **Real economy:** GDP growth, industrial production, consumer confidence, PMI (manufacturing)
  - **External:** Trade balance, commodity prices (oil, metals), capital flows
  - **Institutional:** World Bank governance indicators, central bank independence index, inflation targeting adoption dates

Why this works: Cross-country panel structure, multiple outcomes test mechanism (quantity vs. price effects), policy relevance for international institutions

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Key Principle: More Data = Better Project

Don't limit yourself to 3-5 variables. Professional analyses use 10-20+ variables because:

- **Robustness:** Does your result hold when adding controls?
- **Mechanisms:** Can you test WHY the effect exists?
- **Multiple outcomes:** Does the driver affect returns AND volatility AND volume? (Stronger story)
- **Heterogeneity:** Rich data lets you slice by sector, size, region, time period

Example: A REIT project with only `ret ~ FEDFUNDS` is weak. A project with:

- 4 outcomes (returns, volatility, volume, Sharpe ratio)
- 12 FRED series (rates, inflation, output, labor)
- 3 heterogeneity dimensions (sector, size, leverage)
- Event study component (COVID shock)

...is a **portfolio-quality project** that demonstrates real analytical skills.

Part 2: Effective Data Combinations

The Comprehensive Data Strategy

Core Principle: More Data = Stronger Project

Professional quantitative analyses don't rely on 3-4 variables. They use 15-25 variables because:

- **Robustness checks** require alternative specifications
- **Mechanism testing** needs intermediate outcomes
- **Control variables** rule out confounders
- **Multiple outcomes** show your effect is real (not noise)

Three-Layer Data Structure

Layer 1: Primary Dataset (Your Panel/Time Series) This is where your main outcome variables live.

- **REIT:** Returns, volatility, volume, Sharpe ratios, dividend yields
- **Crypto:** Token returns, trading volume, bid-ask spreads, TVL changes
- **Housing:** Price growth, rent growth, sales volume, price-to-rent ratios
- **Macro:** Employment growth, unemployment, wages, labor force participation

Test multiple outcomes to show robustness. If interest rates only affect returns but not volatility or volume, the effect might be spurious.

Layer 2: Supplementary Data (10-15+ Series)

Build a rich dataset from these sources:

1. **FRED (Federal Reserve Economic Data)** — fred.stlouisfed.org
    - Access: Free API via `pandas-datareader`
    - **Interest rates:** FEDFUNDS, MORTGAGE30US, DGS10, DGS2 (yield curve)
    - **Inflation:** CPIAUCSL, PCEPI, CPILFESL (core), breakeven inflation
    - **Labor:** UNRATE, PAYEMS, AWHMAN, initial claims
    - **Output:** GDP, INDPRO, RSXFS (retail sales), capacity utilization
    - **Credit/Risk:** BAA-AAA spread, VIXCLS, TED spread, mortgage delinquency
    - **International:** DXY (dollar index), crude oil, gold prices
2. **Open Dataset Catalog** — `OpenData_rows.csv` (instructor-provided)
    - Housing: Case-Shiller indices, construction permits, housing starts
    - Labor: State/metro employment, wage growth, job openings
    - Climate: Temperature anomalies, drought indices, natural disasters
    - Financial: Equity indices, commodity prices, crypto market data
3. **Policy/Event Data (Hand-Coded or Scraped)**
    - Fed announcements (rate hike dates, QE start/end)
    - Regulatory changes (SEC enforcement dates, Dodd-Frank milestones)
    - Fiscal events (stimulus payments, tax changes)
    - Firm events (earnings announcements, M&A, CEO changes)

Layer 3: Entity Characteristics (From Your Primary Data)

Cross-sectional variation for heterogeneity analysis:

- **Size:** Market cap quartiles, total assets, GDP
- **Leverage:** Debt-to-equity, interest coverage
- **Risk:** Beta, volatility, credit rating
- **Performance:** Past returns (momentum), profitability ratios
- **Geography:** Region, urban/rural, coastal/inland
- **Sector/Industry:** Property type, SIC code, token type

Data Combination Patterns

Pattern 1: Panel + Time-Varying Macro (Most Common)

Structure: Entity × Time + Time-varying economic indicators

Example (REIT):

```
Primary:    REIT Master (permno × ym)
Merge with: FRED indicators (ym)
Result:     Each REIT-month has its own return + that month's Fed Funds Rate
```

Example (Crypto):

```
Primary:    Token panel (token_id × date)
Merge with: Regulatory severity (date) + Bitcoin return (date)
Result:     Each token-day has its own return + that day's macro conditions
```

Merge strategy: Left join on time variable (keeps all primary observations)

Pattern 2: Panel + Entity-Specific Shocks

Structure: Entity × Time + Entity-specific treatment timing

Example (DiD setup):

```
Primary:    Firm panel (gvkey × datadate)
Add:        M&A announcement dates (gvkey + event_date) → binary `treated`
variable
Result:     Panel with time-varying treatment indicator
```

Use case: Difference-in-Differences analysis (M3)

Pattern 3: Aggregate Time Series + Multiple Drivers

Structure: Single time series + multiple time-varying predictors

Example (Macro):

```
Primary:    US GDP growth (quarterly)
Merge with: FRED rates + fiscal policy + trade data (all quarterly)
Result:     Time series with GDP + policy variables
```

Use case: Time series regression or ARIMA with exogenous variables (M3 Model B option)

Recommended Supplementary Data by Track (Minimum 10-15 Series)

Track	Primary Dataset	Core Supplementary (8-12 series)	Advanced Extensions (5-10 more)

Track	Primary Dataset	Core Supplementary (8-12 series)	Advanced Extensions (5-10 more)
REIT	REIT Master (ret, mcap, sector, leverage, ffo)	**Rates:** FEDFUNDS, MORTGAGE30US, DGS10, TERMSPREAD **Macro:** CPIAUCSL, UNRATE, GDP, INDPRO **Real estate:** CSUSHPISA, TLRESCONS **Risk:** VIXCLS	**Regional:** Case-Shiller 20 cities **Credit:** BAA-AAA spread, mortgage delinquency **Policy:** COVID lockdown dates, Fed QE announcements **Alternative outcomes:** volatility, volume, dividend growth
Crypto	Token panel (return, mcap, volume, token_type, age)	**Crypto:** Bitcoin return, ETH gas price, stablecoin dominance, DeFi TVL **Macro:** VIX, FEDFUNDS, DXY, gold **Policy:** SEC enforcement dates (hand-coded)	**Market structure:** Exchange listings, DEX vs. CEX volume **Network:** On-chain transaction volume, active addresses **Sentiment:** Twitter mentions, Google Trends **Alternative outcomes:** bid-ask spread, TVL changes
Housing	MSA home prices (price_growth, rent_growth, sales_volume)	**Rates:** MORTGAGE30US, FEDFUNDS, DGS10 **Housing:** Case-Shiller, construction permits, inventory **Economic:** UNRATE, median income, GDP	**Supply:** Saiz elasticity index, zoning restrictiveness **Demand:** Migration flows, millennial homeownership **Credit:** Mortgage denial rates, FHA share **Alternative outcomes:** price-to-rent, foreclosure rates
Macro	Country employment (emp_growth, unemp_rate, LFPR)	**Monetary:** Policy rates (4+ countries), inflation, FX rates **Fiscal:** Govt spending, debt/GDP **Real:** GDP, industrial production, PMI	**Institutional:** Labor flexibility index, union density **External:** Trade balance, commodity prices **Confidence:** Consumer/business surveys **Alternative outcomes:** wages, hours worked, sector employment

Rule of Thumb: Your data table should have 15-25 columns (entity ID, time, outcome(s), 10-20 predictors). If you only have 5-6 columns, you need more data.

Data Quality Checklist (Before Committing)

- ☐ **Comprehensive coverage:** You have 10-15+ supplementary variables (more is better for robustness)
- ☐ **Multiple outcomes tested:** At least 2 outcome variables (e.g., returns + volatility, or price growth + volume)
- ☐ **Time alignment:** Primary and supplementary data use compatible time frequencies (monthly-monthly, daily-daily, or aggregate daily to monthly)
- ☐ **Coverage overlap:** Supplementary data covers your primary data's time range (verify start/end dates match)
- ☐ **No duplicate keys:** Each entity-time combination appears once in primary data
- ☐ **Missing values understood:** Know why data is missing (delisting, new listing, data gap)—document percentages
- ☐ **Units match:** Returns in %, not decimals (or vice versa, but consistent across all variables)
- ☐ **Heterogeneity dimensions:** You can slice data by at least 2 groupings (sector + size, or region + time period)

Part 3: GitHub Repository Setup and Workflow

Why GitHub is Your Source of Truth

Problem without version control:

- "Final_v3_REAL_THIS_ONE.py" chaos
- Lost work when files are accidentally deleted
- Can't merge contributions from 4 teammates
- No history of who changed what

Solution with GitHub:

- Single shared repository with full history
- Branching for experimental work
- Version control = rewind time if needed
- Professional workflow for your portfolio

Initial Setup (Week 1-4)

Step 1: Create Team Repository via GitHub Classroom (REQUIRED)

All teams MUST use GitHub Classroom:

1. Accept the GitHub Classroom assignment link (posted in Blackboard)
2. **First teammate:** Create a new team and give it a descriptive name
   - Example: `team-reit-wizards`, `team-crypto-analysts`
3. **Remaining teammates:** Join the existing team (don't create a new one!)
4. GitHub Classroom automatically creates your repository: `qm2023-capstone-[team-name]`

Important: This repository is shared by all team members. Everyone has push access.

Step 2: Open Repository in GitHub Codespaces

What is GitHub Codespaces? A cloud-based development environment that runs in your browser. No local setup needed—Python, Git, and all tools are pre-configured.

How to access:

1. Go to your team's repository on GitHub.com
2. Click the **Code** button (green button)
3. Click the **Codespaces** tab
4. Click **Create codespace on main**
5. Wait 30-60 seconds for environment to load
6. You're now in a VS Code-like editor in your browser!

Each teammate can:

- Open their own Codespace (you can have multiple Codespaces for the same repo)
- Work independently in the cloud
- Commit and push changes back to the shared repository

Step 3: Set Up Project Structure (One teammate does this first)

Create folders (in Codespaces terminal):

```
mkdir -p code data/raw data/processed data/final results/figures results/tables
results/reports tests
```

Add essential files:

1. `README.md` (update the template provided):

    ```
    # QM 2023 Capstone Project: [Team Name]

    **Team Members:** [List names]
    **Dataset:** [REIT Master / Alternative: X]
    **Research Question:** [Your question]

    ## Project Structure
    See `Course-Design/Capstone-Project/STRUCTURE.md` for folder guide.

    ## Setup in Codespaces
    ```bash
    pip install -r requirements.txt
    python code/config_paths.py  # Verify paths
    ```
    ```

2. `code/config_paths.py` (copy from `QM-2023-Capstone-Repo/code/config_paths.py` in Course-Design folder)

3. `requirements.txt`:

```
pandas>=2.0.0
numpy>=1.24.0
matplotlib>=3.7.0
seaborn>=0.12.0
pandas-datareader>=0.10.0
statsmodels>=0.14.0
linearmodels>=5.3
scikit-learn>=1.3.0
rich>=13.0.0
```

4. `.gitignore`:

```
# Python
__pycache__/
*.pyc
.env

# Data (don't commit large files)
data/raw/*.csv
data/processed/*.csv

# Keep final output (small, important)
!data/final/*.csv

# Jupyter
.ipynb_checkpoints/

# OS
.DS_Store
Thumbs.db
```

Commit and push (in Codespaces terminal):

```
git add .
git commit -m "Initial project structure"
git push origin main
```

Other teammates: When you open your Codespace, you'll see these updates automatically. If not, use the Source Control panel in Codespaces (left sidebar) and click "Sync Changes".

Collaborative Workflow (Weeks 4-15)

The Main ↔ Feature Branch Pattern

Main branch: Stable, working code. Always runs without errors.

Feature branches: Experimental work. Create for each milestone or major task.

Workflow (in GitHub Codespaces):

1. **Before starting new work, sync with latest changes:**
   - Click the **Source Control** icon in the left sidebar (looks like a branching diagram)
   - Click **Sync Changes** to pull the latest from GitHub
   - Or use terminal: `git checkout main && git pull origin main`
2. **Create feature branch:**

   ```
   # Example: Working on M1 data cleaning
   git checkout -b m1-data-cleaning
   ```

   Or use the branch selector in the bottom-left corner of Codespaces → **Create new branch**
3. **Do your work, commit often:**
   - Make changes to your files
   - Click **Source Control** icon (left sidebar)
   - You'll see changed files listed
   - Stage files by clicking the **+** icon next to them
   - Enter a commit message in the text box (e.g., "M1: Handle missing returns by dropping rows")
   - Click **Commit** (checkmark icon)
   - Click **Sync Changes** to push to GitHub

   Or via terminal:

   ```
   git add code/capstone_data_pipeline.py
   git commit -m "M1: Handle missing returns by dropping rows"
   git push origin m1-data-cleaning
   ```
4. **When feature is done and tested, merge to main:**
   - Switch to main branch (branch selector, bottom-left)
   - Sync changes to get latest: `git pull origin main`
   - Merge your feature branch: `git merge m1-data-cleaning`
   - Push updated main: `git push origin main`
5. **Delete feature branch (cleanup):**

```
git branch -d m1-data-cleaning
```

Handling Merge Conflicts (Don't Panic)

When they happen: Two teammates edit the same lines of the same file.

How to resolve:

1. Git will show:

```
CONFLICT (content): Merge conflict in code/capstone_data_pipeline.py
```

2. Open the file. You'll see:

```
<<<<<<< HEAD
# Your version
df = df.dropna(subset=['ret'])
=======
# Teammate's version
df = df.dropna(subset=['ret', 'mcap'])
>>>>>>> m1-data-cleaning
```

3. **Decide which to keep** (or combine both):

```
# Combined version
df = df.dropna(subset=['ret', 'mcap'])
```

4. **Remove conflict markers** (`<<<<<<<`, `=======`, `>>>>>>>`), save, commit:

```
git add code/capstone_data_pipeline.py
git commit -m "Resolve merge conflict: keep both ret and mcap in dropna"
git push origin main
```

Prevention: Communicate who's working on what. Use different files or feature branches.

File Organization Best Practices

What to commit to GitHub:

☑ **Always commit:**

- All code (`.py`, `.ipynb`)
- Small output files (`data/final/*.csv` if <10 MB)
- Documentation (`.md` files, `AI_AUDIT_APPENDIX.md`)
- Figures (`results/figures/*.png`)
- Tables (`results/tables/*.csv`)
- `requirements.txt`, `README.md`, `.gitignore`

❌ **Never commit:**

- Large raw data files (>10 MB) — use `.gitignore`
- Virtual environments (`.venv/`)
- Passwords or API keys (use `.env` file, add to `.gitignore`)
- Temporary files (`__pycache__`, `.ipynb_checkpoints`)

Why: Keep repo size manageable, protect secrets, maintain reproducibility.

Milestone Submissions (M1-M4)

Structure your commits (in Codespaces):

1. Stage all milestone files using Source Control panel
2. Or via terminal:

```
# M1 Example
git add code/capstone_data_pipeline.py
git add data/final/reit_analysis_panel.csv
git add data/final/reit_metadata.json
git add M1_data_quality_report.md
git add AI_AUDIT_APPENDIX.md
git commit -m "M1 submission: Data Pipeline complete"
git push origin main
```

Tag milestones for easy reference:

```
git tag -a M1-submission -m "Milestone 1: Data Pipeline"
git push origin M1-submission
```

Later, you can return to this exact version:

```
git checkout M1-submission
```

Team Collaboration Norms

Daily/Weekly Git Hygiene:

1. **Sync before you work:** Avoid conflicts
    - Open Codespaces → Click **Sync Changes** in Source Control panel
    - Or terminal: `git pull origin main`
2. **Commit messages:** Descriptive, present tense
    - Good: `"Add FRED API integration for M1"`
    - Bad: `"stuff"`, `"fixed it"`, `"asdfasdf"`
3. **Commit often:** Small, logical chunks
    - After fixing a bug
    - After completing a function
    - Before switching tasks
    - **Not:** End-of-day dump of 10 changes
4. **Use branches for experiments:**
    - Testing a new cleaning method? → Create branch `test-winsorize`
    - Works? Merge to main. Doesn't work? Delete branch, no harm done.

Code review before milestone submission:

- Assign one teammate as "Release Manager" per milestone
- That person does final check: code runs, paths are relative, files organized
- Tag and push final submission
- All teammates verify by opening a fresh Codespace and running the script

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Contribution Tracking (Required for M4)

Simple log in `team_contributions.md`:

```
# Team Contribution Log

## Week 4 (Feb 9-13): M1 Planning
- Sarah: Set up GitHub repo, created folder structure (2 hours)
- Alex: Downloaded REIT data, explored with pandas (2 hours)
- Jordan: Researched FRED API, tested pandas-datareader (1.5 hours)
- Taylor: Drafted M1 outline, listed cleaning decisions (1 hour)

## Weeks 5-6 (Feb 16-25): M1 Execution
- Sarah: Wrote data cleaning section, handled missing values (3 hours)
- Alex: Integrated FRED API, merged with REIT data (2.5 hours)
- Jordan: Validated merge, verified row counts (1 hour)
- Taylor: Wrote M1_data_quality_report.md, final testing (2 hours)
```

Update weekly. This makes M4 individual addendum trivial to write.

Common Git Issues and Fixes

Issue 1: "I forgot to sync before making changes"

In Codespaces:

1. Click **Source Control** → **Sync Changes** (if conflicts appear, see merge conflicts section)

Via terminal:

```
git stash  # Save your changes temporarily
git pull origin main  # Get teammate's changes
git stash pop  # Reapply your changes
# Resolve conflicts if any, then commit
```

Issue 2: "I committed sensitive data (password, API key)"

```
# Remove file from Git history (before anyone else pulls!)
git rm --cached code/my_secrets.py
git commit -m "Remove secrets file"

# Add to .gitignore
echo "code/my_secrets.py" >> .gitignore
git add .gitignore
git commit -m "Add secrets to gitignore"
git push origin main
```

Prevention: Use `.env` files for secrets, add `.env` to `.gitignore` from day 1.

Issue 3: "I want to undo my last commit"

```
# Undo commit but keep changes (most common)
git reset --soft HEAD~1

# Undo commit and discard changes (careful!)
git reset --hard HEAD~1
```

Issue 4: "My teammate's branch has changes I need"

```
# Fetch their branch
git fetch origin
```

```
# Check out their branch
git checkout their-branch-name

# Or merge their branch into yours
git checkout your-branch-name
git merge their-branch-name
```

In Codespaces: Use the branch selector (bottom-left) to switch between branches.

Part 4: Putting It All Together

Week-by-Week Checklist

Week 1-3: Exploration

- ☐ Form team (3-4 students)
- ☐ Set up team GitHub repository
- ☐ Clone repo to all local machines
- ☐ Decide: Default REIT track OR alternative dataset?
- ☐ If alternative: Draft proposal (due Fri, Week 4)
- ☐ Browse supplementary data options (FRED, Open Dataset Catalog)

Week 4: Finalization

- ☐ Submit team roster (Mon, Feb 9)
- ☐ Submit alternative dataset proposal (Fri, Feb 13) — if applicable
- ☐ Hold first team meeting: assign M1 roles
- ☐ Download primary dataset, explore structure
- ☐ Identify 3-5 supplementary variables to integrate

Weeks 5-6: M1 Execution

- ☐ Write `capstone_data_pipeline.py`
- ☐ Clean primary dataset (missing values, outliers, duplicates)
- ☐ Fetch supplementary data (FRED API or manual download)
- ☐ Merge datasets on time variable
- ☐ Save `[dataset]_analysis_panel.csv` and `metadata.json`
- ☐ Write `M1_data_quality_report.md`
- ☐ Update `AI_AUDIT_APPENDIX.md`
- ☐ Test reproducibility (delete `data/final/`, rerun script)
- ☐ Commit, tag `M1-submission`, push to GitHub
- ☐ Submit by Wed, Feb 25, 11:59 PM

Key Success Factors

1. **Start early:** Don't wait until the deadline week
2. **Communicate constantly:** Daily Slack/Discord updates

3. **Use GitHub properly:** Commit often, pull before push, branch for experiments
4. **Document everything:** Contribution log, AI Audit, data decisions
5. **Test reproducibility:** If your script doesn't run on a teammate's machine, it won't run for the instructor

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Resources

- **Main Capstone README:** `Course-Design/Capstone-Project/README.md`
- **Dataset Translation Guide:** `Course-Design/Capstone-Project/Dataset-Translation-Examples.md`
- **Alternative Dataset Catalog:** `OpenData_rows.csv` (instructor-provided)
- **M1 Detailed Spec:** `Course-Design/Capstone-Project/M1-Data-Pipeline/README.md`
- **Team Formation Guide:** `Course-Design/Capstone-Project/team_formation.md`
- **Office Hours:** Dr. Seagraves, Mon & Wed 3-5 PM (Helm 122-D)

FAQ

Q: Can we use data not in the Open Dataset Catalog? A: Yes, but requires proposal and approval. Must meet minimum requirements (50+ entities, 5+ years, reproducible access).

Q: How many supplementary datasets should we add? A: Aim for 10-15 series minimum. More is better for robustness—professional projects use 20+ variables. Quality AND quantity matter. Don't just pick 3 variables and stop; richer data = stronger analysis.

Q: What if our alternative dataset proposal is denied? A: You default to REIT Master track. No penalty. Approvals are based on feasibility, not creativity.

Q: Can we change our research question after M1? A: Minor refinements? Yes. Major pivot? Consult instructor. M2 EDA often reveals better angles.

Q: Do we need separate GitHub repos for weekly assignments and capstone? A: No! Use GitHub Classroom repos for weekly assignments (auto-graded). One shared team repo for capstone (all milestones M1-M4).

Q: What if a teammate doesn't contribute to GitHub commits? A: GitHub contribution graph is visible. Peer evaluations (Week 15) will factor in documented contributions. Contact instructor early if issues arise.

This is your roadmap. Follow it, and you'll have a portfolio-quality project by May.

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